

Eskom's funding crisis will spark new energy deal

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Eskom is facing an acute funding crisis which is likely to persist into the next decade. As climate-friendly technology options are considered, its generation mix will become even more capital-intensive, making it more difficult to finance these off its balance sheet. The combination of Eskom's financial woes and pressures to mitigate climate change will more than likely lead to a new electricity market structure in South Africa where the private sector plays a much bigger role.

Eskom has just made a R12,98-billion, before tax, loss. Over the next five years its capital expansion programme will approach R400 billion (nearly ten times the budget for the arms deal). Despite government assistance, in the form of a R60 billion long-term subordinated loan and debt guarantees totalling R177 billion, Eskom faces an R80 billion funding gap.

In the past, Eskom funded its investment programme through private debt and retained earnings. Adequate tariff levels made this possible. This year, inexplicably, Eskom ducked the imperative of applying for increases sufficient to fund investment in new capacity. It is mind-boggling that Eskom is in the middle of a massive multi-year investment programme without any clarity on its funding model.

What is to be done? There are limits on how much Eskom can borrow, although Eskom is being more aggressive in accessing capital markets through a range of financing instruments, including local and international bonds, export credit, and loans from development finance institutions such as the World Bank and the African Development Bank.

It is unlikely that government can provide more in terms of equity for Eskom, given the recession, reduced tax income and increased demands for social expenditure. Perhaps a special tax-payer levy, further sub-ordinated loans or sovereign guarantees can be explored, but none of these will be easy in the current economic context. Ultimately debt has to be repaid and there is only one sustainable source of revenue – electricity tariffs. As Public Enterprises Minister, Barbara Hogan, made clear in her recent budget speech, prices need to cover costs, including the cost of capital. Steep tariff hikes are thus likely to continue into the future.

In addition to these financing challenges, Eskom will have to start confronting the imperative of mitigating climate change. More than 90% of South Africa's electricity is generated from coal. South Africa and other emerging economies argue that they still need to invest in least-cost power generation technologies in order to grow their economies and meet socio-economic needs. Rather than limiting emissions at below those of industrialised countries they seek a more just outcome where the world moves to equal emission rights.

The problem is that South Africa's carbon dioxide emissions are already way above the global average in both per capita terms.

South Africa's wiggle room regarding greenhouse gas emissions is diminishing. Last year, Cabinet noted the recommendations from the Long Term Mitigation Scenario modelling exercise which would permit South Africa's emissions rising to a plateau in the early 2020s and then declining. A policy process was announced in March this

year which aims to produce a White Paper on a National Climate Change Response Policy by the end of 2010.

Eskom's Board are aware of these developments and are considering alternative investment paths which would limit carbon emissions after 2025. Such a scenario almost certainly includes building a third mega coal-fired power station after the two currently being constructed – Medupi and Kusile – and, perhaps, also a fourth.

The big question is what will replace coal? Front runners in planning scenarios are nuclear energy and concentrated solar plants (CSP). But, as intimated earlier, these technologies are much more capital intensive than coal and will cost two to three times more per installed megawatt. Indeed CSP, with its lower capacity factor, will be even more expensive per effective megawatt. Wind energy will also play a role, but not as large as the other base-load technologies.

And so we are back to the challenge of how we shall finance future power generation capacity. Eskom's funding crunch is most acute over the next three years. Large tariff increases will help. But the next wave of investment decisions, that will soon have to be made, will once again create a financing crisis which will probably only be solved by facilitating the entry of more players into the electricity industry.

This is already government policy: Cabinet decided long ago that up to 30% of new power generation could come from private independent power producers (IPPs).

Eskom has not yet signed any power purchase agreements with IPPs. That will have to change.

Electricity sector planning has now been shifted to Eskom's System Operator who is producing a national plan, including low carbon scenarios. These will soon be published. What is still missing is a governance mechanism which will encourage more transparency and confidence in the impartiality of the planning process.

The Department of Energy is talking about an Independent System Operator which could create a more objective institutional base for contracting the private sector. That would involve a profound restructuring of Eskom and might not happen in the short term. In the meantime, the regulator is trying to implement feed-in tariffs for renewable energy technologies, but much work still needs to be done on acceptable power purchase and transmission access agreements.

The issue of power sector market restructuring is thus once again on the table. It is clear that Eskom cannot do everything, nor does it wish to. The associated pressures of climate change mitigation and financing will gradually erode barriers to private sector entry and South Africa's power sector will migrate inevitably to a market structure that encompasses more diversity and cleaner technologies.

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